

Domain-Specific Small Language Model for Medical Advice

Data Card

Team Tana - TRI AI Saturdays, Cohort 10

Project	Domain-Specific SLM for Medical Advice
Team	Tana
Cohort	TRI AI Saturdays, Cohort 10
Challenge	01 Build Your Own Small Language Model, Data Card
Final Training Set	141 pairs (43 competition-provided + 98 team-authored)

Input vs. Output

Users input a natural-language patient- or frontline-worker-style question (text), e.g. “BP readings improved, skip tablets this weekend?”. At inference time this question is paired with a retrieved context passage, the text of one of the competition's 24 public-health factsheets, selected via metadata filtering (topic, care_setting, population) and embedding similarity.

The model generates a short (1–3 sentence) text answer in a direct, evidence-grounded style, including an explicit action or triage cue where the source content indicates urgency. This is a generative text-to-text task, not a fixed-label prediction.

Classification Problem

There is no primary classification task; the core task is generative (question → short answer). However, every example carries three categorical metadata labels used for retrieval filtering.

Topic (8 classes): chronic_disease, infectious_disease, maternal_health, medication_safety, nutrition, vaccination, emergency_triage, mental_health_basics. Care setting (5 classes): primary_care, community, hospital, home, school.

Population (5 classes): adult, child, infant, pregnant, general. These categories were chosen because they mirror how a frontline health worker actually triages a question in practice by subject area, by where care is delivered, and by who the patient is rather than by formal clinical taxonomy, since our end users are non-specialist community and frontline workers.

Labeling

The original 24 factsheets, 43 question–answer pairs, and their topic/care_setting/population labels were provided by the competition organizers. Team Tana authored 98 additional question–answer pairs directly (not crowd-sourced or AI-generated), inheriting the existing labels from each pair's linked source factsheet rather than creating new label categories.

Every team-authored pair was written by a team member following a shared rubric: the answer must be traceable entirely to its linked factsheet's text, phrased the way a real patient or frontline worker would ask it, and matched to our target answer style (1–3 sentences, direct verdict opening, no hedging language). A second team member spot-reviewed a sample of pairs against this rubric before merging them into the training set.

Data Creation

We did not collect data from interviews, recordings, or the web. Our dataset combines the competition-provided synthetic corpus (24 factsheets, 43 QA pairs) with 98 additional QA pairs written manually by team members, each grounded strictly in the same 24 factsheets.

Synthesized but human-authored synthesis, not AI-generated or scraped. We deliberately chose manual authorship over AI-generated augmentation to maintain a clear, auditable chain of accountability for every training example, given the safety stakes of clinical guidance content.

Note: we also explored augmenting with real, externally cited WHO/CDC sources, but found it altered our model's answer style away from the competition's expected format, so it was excluded from our final submitted training set. See Values and Community Alignment below.

Web Collection Ethics

No web scraping was used in our final training set. All content originates from the competition-provided factsheets or team-authored text derived from them.

Human team members authored all additional training pairs directly, without AI-generation tools, to preserve a clear authorship and accountability trail.

Not applicable to our submitted dataset, no identifiable individuals, real patient records, or third-party web content are represented anywhere in it.

Bias and Representation

Because our factsheets are synthetic and English-language, the dataset doesn't capture real-world variation in symptom description, local terminology, dialect, or multilingual phrasing that a real deployment would need to handle. Our team's own framing of “patient-style” questions also reflects our assumptions about how patients ask questions, which may not generalize across the full diversity of communities this tool is meant to serve.

We deliberately stratified our 98 team-authored pairs across all 8 topics and all care_setting/population labels rather than concentrating on the easiest cases, so the dataset doesn't over-represent any single population group (e.g. general adult questions) at the expense of others (e.g. pregnant, infant, or child-specific guidance).

Values and Community Alignment

Our problem statement identified safety, groundedness, and accessibility as core values. Safety shaped our decision to validate every team-authored answer against its source factsheet rather than allow fabricated medical claims into training data. Groundedness shaped our retrieval-linked architecture, every training pair carries a document_id, so every model answer is traceable to a specific, reviewable source. Accessibility shaped our insistence on short, direct, jargon-free phrasing, since our target users are frontline workers needing fast, actionable guidance rather than clinical-register explanations.

Because every pair is traceable to its source document, any future error in model output can be diagnosed as either a factsheet gap or a training-data quality issue, rather than remaining an unexplainable failure, a transparency mechanism we consider essential for any tool touching clinical guidance, even at this synthetic, benchmark stage. We also tested, and deliberately excluded, additional real-world sourced data

once we found it compromised answer consistency, choosing benchmark reliability over unvalidated scope expansion, a decision we see as itself reflecting our safety-first value.